MATH 114 HW 3

Dani Nguyen

Winter 2025

Question 1

Let $k = 1$ be the class of companies that issue dividends and $k = 2$ be the class of companies that didn't.
Let X_1 be the percent profit of $k = 1$ and X_2 be the percent profit of $k = 2$
We have

$$X_1 \sim N(10, 36) \qquad\qquad X_2 \sim N(0, 36)$$

$$P(k = 1) = .8$$

$$\begin{aligned}
P(k = 1|X) &= \frac{P(y = 1)P(X|y = 1)}{P(y = 1)P(X|y = 1) + P(y = 2)P(X|y = 2)} \\
&= \frac{.8f_1(X)}{.8f_1(X) + .2f_2(X)} \\
&= \frac{.8\frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_1)^2)}{.8\frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_1)^2) + .2\frac{1}{\sqrt{2\pi\sigma^2}}\exp(-\frac{1}{2\sigma^2}(X-\mu_2)^2)} \\
&= \frac{.8\exp(-\frac{1}{2\cdot 36}(4-10)^2)}{.8\exp(-\frac{1}{2\cdot 36}(4-10)^2) + .2\exp(-\frac{1}{2\cdot 36}(4-0)^2)} \\
&= \frac{.8\exp(-\frac{1}{72}(-6)^2)}{.8\exp(-\frac{1}{72}(-6)^2) + .2\exp(-\frac{1}{72}(4)^2)} \\
&= \frac{.8\exp(-1/2)}{.8\exp(-1/2) + .2\exp(-2/9)} \\
&= 0.7519
\end{aligned}$$

Question 2
When K=1, KNN fit perfectly interpolates the training observation which gives zero error on the training data set. This means the test error rate is 36%. Therefore, logistic regression is better.

Question 3

a. We split our data according to k subsets, and then we save each k'th set for testing, and training on the remaining $k-1$ sets combined.

b. A drawback of the validation set approach is that it is very sensitive to which data is included in the training, it is however less computationally expensive because we only have to fit one model. For the LOOCV, we set $k = n$ and use most of the data for training, saving one observation for validation. This is very computationally expensive compared to validation sets, but has far less bias.

Question 4
Question 5

1.